

LAB 2 – TRASHTAG PRODUCT SPECIFICATION

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1 Introduction

Litter and illegal dumping are common problems all people have seen. Litter is not just unsightly; it is dangerous and directly impacts the environment and all of those living in it, including humans.

Improperly discarded waste accounts for an estimated 100,000 marine animal deaths annually (Texas Disposal Systems 2024). According to a survey from 2016 roughly 21% of beachgoers report they have been injured from litter such as glass or metal from bottles and cans. The danger and ugliness of litter drives people away. The City of Hampton reports a drop in property values of 7-10% in areas with illegal dumping. Tourist destinations with poor litter control and cleaning see drops in revenue up to 38% (Sorensen 2024). Furthermore, when people see litter, they are more likely to add to it instead of finding proper disposal locations (Blouin 2016).

TrashTag aims to improve the litter and waste cleanup process by developing a mobile application where users can photograph and report piles of waste, give their exact locations, and connect with cleanup groups that are well-equipped to handle the cleanup process in order to improve environmental wellbeing and preserve nature's beauty for the public eye.

1.1 Purpose

This document serves as the guide for developers to build the TrashTag application and any necessary support applications or content. It explains each feature of the application and that feature's purpose.

Contained herein is also a description of any external components necessary for the function of this application. Stakeholders may also make use of this document as a reference for intended functionality of the application.

1.2 Scope

TrashTag is a web application to connect everyday citizens with environmental clean-up organizations and local governments to more quickly clean large litter and bulky dumped items. Features that revolve around finding and reporting trash have an interface focusing on mobile use and features that revolve around planning and analytics assume the user is on a desktop or large screen device. All features are usable in either context.

The core functionality of TrashTag is a live map of reported bulk litter and dumping. Users can report litter and attach photos to the report. The map lets users see which litter is nearby to their location or see reports in an area of interest. Those reports show as items on the map for other users who can then see the full details. The images on the reports allow for clean-up crews to accurately gauge the scope of the litter and plan accordingly.

Users can indicate in reports if the trash is still there or if they have cleaned it. Which users have reported and which users have cleaned is stored along with the report. TrashTag uses that data to maintain a ranking system for users based on how much they have reported and cleaned. Users can also update their own reports to add additional information.

When TrashTag detects a user is nearby reported trash, the user is given the option of verifying that it still is there. If a user has indicated they have cleaned previously reported trash, nearby users are requested to verify the trash is gone.

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1.3 Definitions, Acronyms, and Abbreviations

Analytics: The collection of data sent to local government authorities and/or organizations providing them with snapshots of data regarding littering in specified areas.

API: Application Programming Interface. A software mechanism of accepting and providing data from and to external applications.

Dumping/Trash/Litter: Any material purposefully or otherwise left in an area where it does not belong.

Environmental Organization: Private, and frequently non-profit, groups dedicated to environmental causes.

Geotag/Geotagging: GPS provided location information embedded into images taken with a camera.

Report/Reporting: The set of information and photos that users individually enter in the application to indicate illegally dumped trash they have found in the public.

User: Any individual that makes use of the application.

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1.5 Overview

The remainder of this document contains the overall description of the product as well as the detailed list of features, characteristics, and constraints of the application. Section 2 goes into detail of features of TrashTag in addition to the major functional components, user interfaces, and intended user characteristics. Section 3 defines ... *TBD*.

2 Overall Description

2.1 Product Perspective

TrashTag has big goals, but the implementation is relatively simple. The client front end is built in React and provides the map interface using OpenStreetMap. This is a web app designed to display on mobile or desktop easily thus making native mobile apps unnecessary. The back end is handled with Python and Django. For the prototype, the data is stored in a basic SQLite database. However, through the power of object orientation and the Django Object Relational Model, the database can easily be switched with a more powerful one such as MySQL. Digital Ocean droplets provide powerful, flexible, and affordable co-hosted virtual machines that can be expanded easily to handle additional users. **Figure 1** *Major Functional Components Diagram* (below) shows how each of these pieces related and connect to one another.

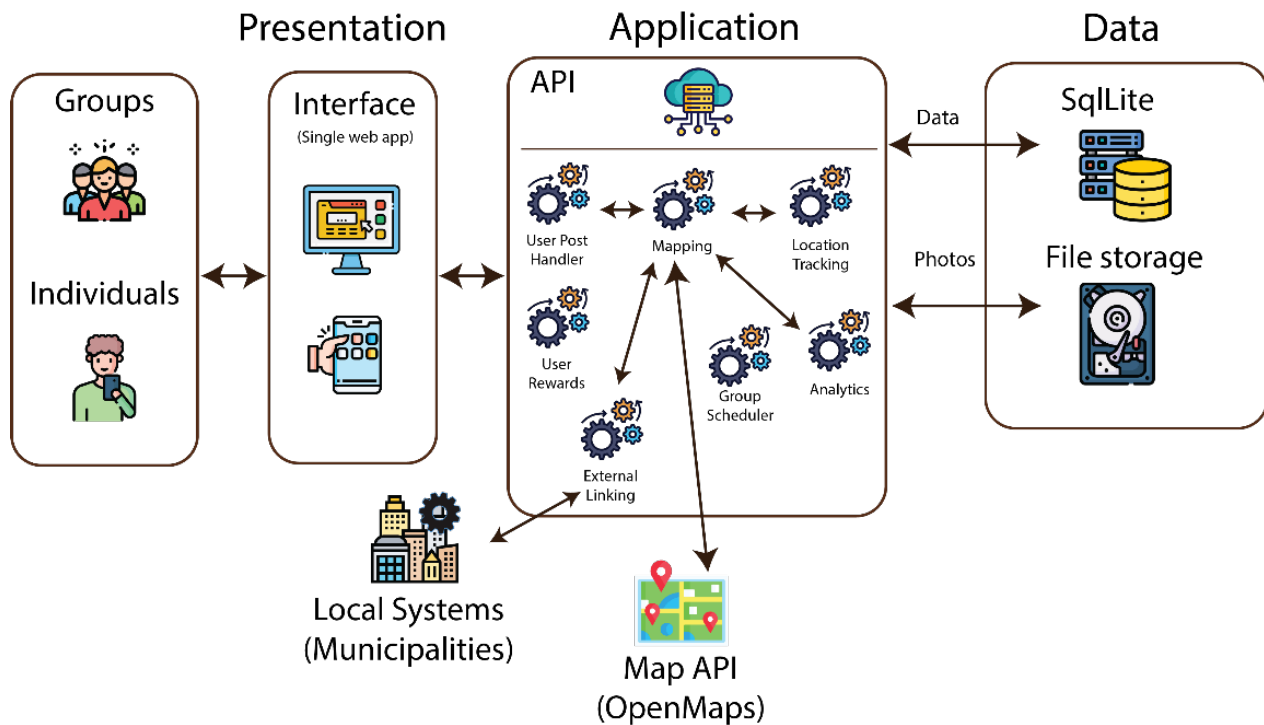


Figure 1 Major Functional Component Design

2.2 Product Functions

Nearly all of the real world product features will be implemented in the prototype version of TrashTag. Table 1 below contains all the features and indicators of which will be implemented in the prototype versus the real world product.

Feature	In Prototype?
Litter Reporting	
Report Litter Location	Yes
Upload Photos	Yes
Mark Trash Size (lbs)	Yes
Cleanup Verification status	Yes
Mapping	
View Litter Map	Yes
Litter Density Map	No

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Filter By Status	No
Filter By Location	Yes
Cleanup Tracking	
Update Cleanup Status	Yes
Upload Cleanup Proof	Yes
Monitor Cleanup Status	Yes
Verification	
Verify Accurate Cleanup Status	Yes
Verify Cleanup Completion	Yes
Flag Suspicious Content	No
Analytics	
Highlight Predicted Litter Trends	No
Download Reports	Yes
View Density - Heavy Litter Analytics	Yes
Track Event Impact	No
View Cleanup Statistics	Yes
Event Coordination	
Coordinate Cleanup Events	No
Claim Cleanup Area	No
Delegate Cleanup Tasks	No
Event Participation	
View Cleanup Events	No
RSVP to Events	No
Check-In to Events	No
Proximity Cleanup Alerts	No
Community Impact	
View Impact Statistics	Yes
Gamifications/Rewards	Yes
Top Contributors	Yes

Resources	
Local laws and trash resources	No
Map Embedding	
Embed Map Widget	Yes (basic)
Platform Moderation	
Review Flagged Reports	No
Respond to Flagged Reports	No
Edit/Correct Flagged Report Data	No
Remove Suspicious Reports	No
Mark Reports as Spam	No
Issue Violation Warnings	No
Client Authentication and Authorization	
Create New Account	Yes
Login	Yes
View and Edit Profile	Yes

Features not implemented in the prototype are deemed too time-consuming for the current deadline or depend on another feature that will not be implemented. The prototype focuses on core functionality and the features that most differentiate from competing products.

2.3 User Characteristics

The users of TrashTag include everyday citizens that see large, dumped items in inappropriate areas, non-profit environmental organizations, and local governments. All users are expected to have basic mobile phone usage skills. Environmental organizations and local government employees are expected to have computer skills that allow them to download and view analytic reports in any available file formats.

A typical use scenario involves an everyday citizen capturing a photo of bulk dumped trash and uploading that photo with the TrashTag application. Other users or representatives of environmental

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organizations or local governments can view that report on a map and create a plan for hauling that trash, and they then can mark the report as clean in the TrashTag application.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

TrashTag depends on the open source and freely usable OpenMaps API for map generation and display. The TrashTag API is built with the Django framework in Python and the frontend is built with React framework in JavaScript. TrashTag depends on these projects for security updates and maintenance.